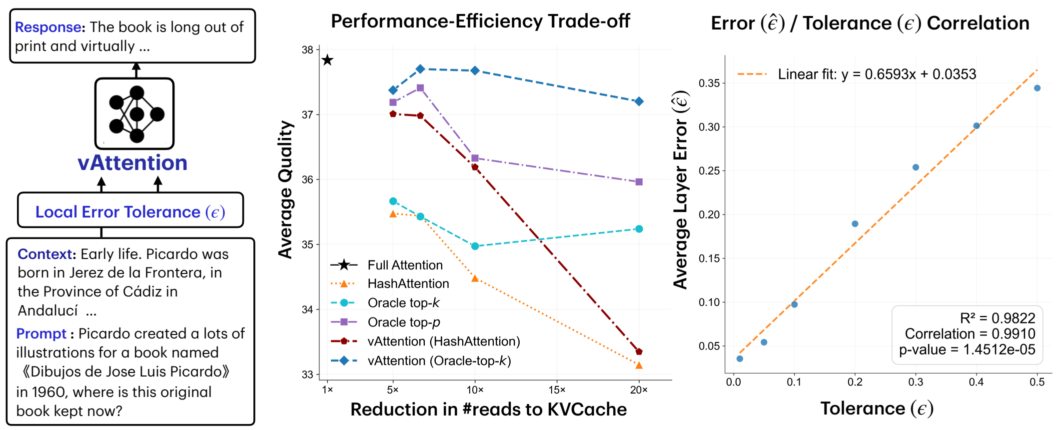

Figure 1: [**Left**:] vAttention accepts user tolerance parameter ϵ and ensures that sparse attention errors are controlled to be within this tolerance. [**Middle**] vAttention achieves a SOTA trade-off, outperforming leading methods like HashAttention and even a strong oracle top-p on a mix from long-context benchmarks (RULER32K, LongBench, Loogle). [**Right**] There is a strong correlation between the approximation error in the layer attention output and the user-defined parameter ϵ accepted by vAttention with verified denominator-only approximation, validating the practical relevance of ϵ parameter

inner products $k_i^\top q$, also known as the top-k approach (top-p, its extension, chooses budgets per head). Thus, sparse attention research is dominated by approaches designed to approximate top-k (Xiao et al., 2024; Tang et al., 2024; Desai et al., 2025; Li et al., 2024b; Hooper et al., 2024; Zhang et al., 2025) and top-p (Zhu et al., 2025) efficiently. However, any deterministic sparsity, such as top-k, assumes that the contextual embedding of the current token can be determined by a small set of specific tokens in the context as if information is encoded in discrete units – an assumption that is clearly not true in input layer (e.g. next word does not depend on only a few words) and cannot be reliably assumed in deeper layers. It is not surprising that attention scores are not always sharply distributed, and in such cases, top-k methods fail to give a good approximation (see Fig. 2). While top-k tokens often dominate attention outputs, contextual meaning arises from the entire distribution of key–value vectors across all tokens. This motivates the statistical perspective behind Verified Sparse Attention (vAttention).

Recently, LSH-based sampling was used to approximate attention (Chen et al., 2024). vAttention is inspired by this work. vAttention is based on a key observation for approximating a sum of n terms. If the sum is sharply dominated by a few terms, selecting those terms provides the best approximation. Conversely, if the terms are similar in value, a case in which top-p leads to choosing an unnecessarily large number of terms, a sampling-based estimator can approximate the sum with a small sample. vAttention combines both strategies, and adjusts the sample size to guarantee a user-specified (ϵ, δ) approximation to the target sum. Using this intuition, vAttention approximates both the numerator and denominator to a specified accuracy, ensuring that the overall error in attention output incurs at most an ϵ relative error with probability $(1-\delta)$, thus providing a "verified" sparse attention. To the best of our knowledge, vAttention is the first practical algorithm to provide approximation guarantees while giving users explicit control over the quality–efficiency tradeoff. This not only provides state-of-the-art quality on sparse attention, but it also makes a compelling argument in favor of deploying sparse attention reliably in the wild.

We extensively evaluate vAttention across diverse models (Llama-3.1-8B-Instruct, Deepseek-R1-Distill-Llama-8B, Mistral-7B-Instruct-v0.3) and benchmarks (RULER (Hsieh et al., 2024), Long-Bench (Bai et al., 2024), Loogle (Li et al., 2023), AIME (Maxwell-Jia, 2024)) and composing it with oracle-top-k, and HashAttention (Desai et al., 2025), a state-of-the-art approximate top-k. We find that vAttention consistently achieves higher accuracy than top-k methods and delivers a superior quality–efficiency trade-off, surpassing even the strongest oracle top-p baseline (See Figure 1. For example, at 10% sparsity, vAttention combined with HashAttention improves RULER32K-HARD accuracy by upto 4.5 percentage points over HashAttention across models. Furthermore, owing to its